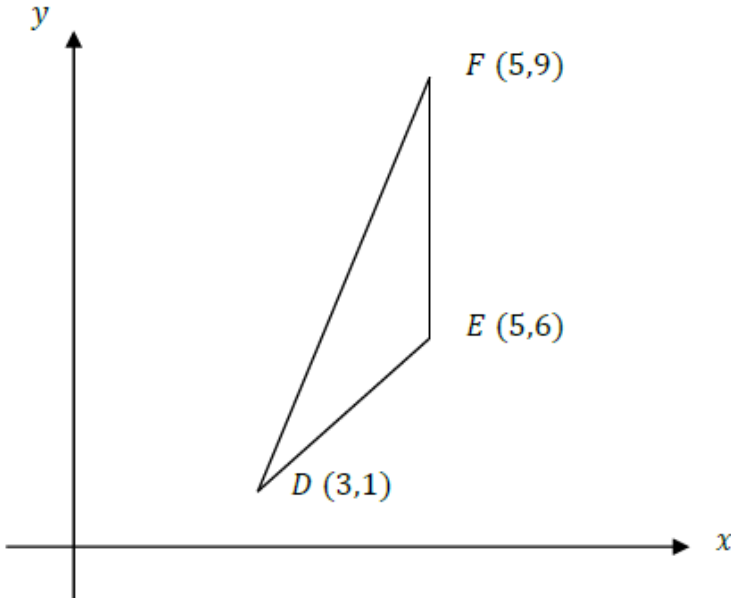
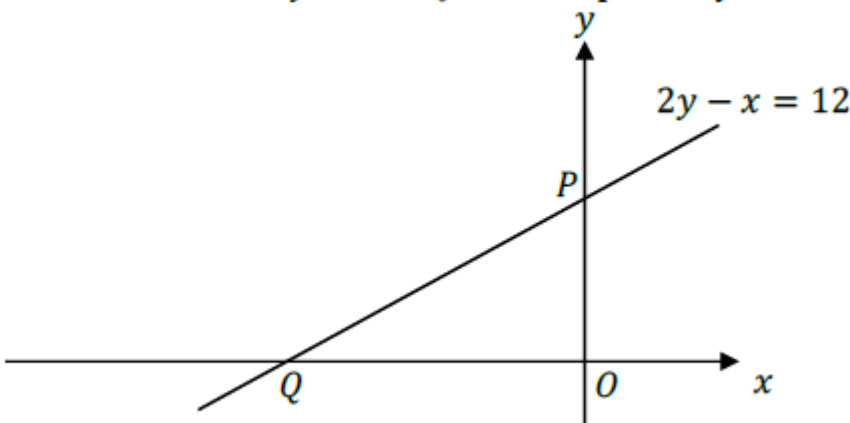
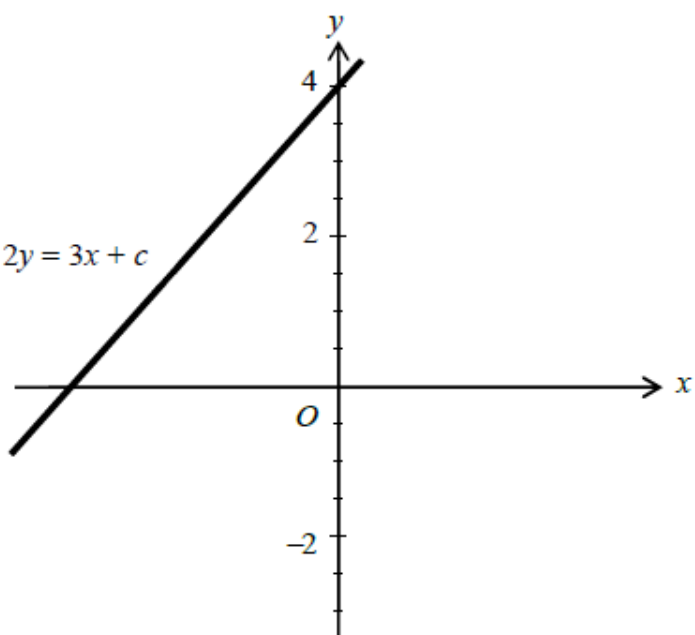
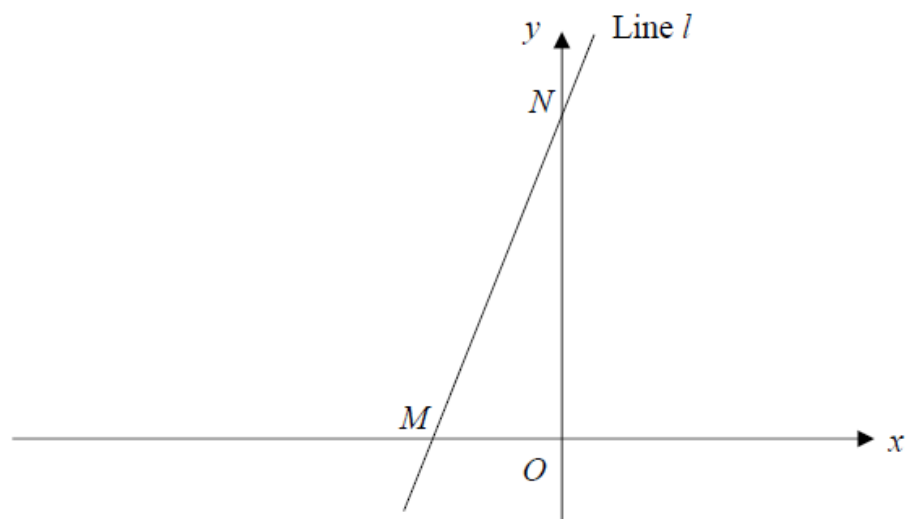


1	YTSS/2025/I/19
	(a) S is a point on the x-axis such that it is 2 units away from $(5, -2)$. Find the coordinate of S. [2] (b) S is a point on the line l_1 which has a y-intercept of -10. Line l_2 has a gradient of 2. James claims that l_1 and l_2 will never meet. Do you agree with his claim? [2]
2	SST/2025/I/14
	The diagram shows the points $D (3, 1)$, $E (5, 6)$ and $F (5, 9)$.  (a) Find the coordinates of G such that $DEFG$ is a parallelogram. [1] (b) Find the length of the line DF [1]

3	SST/2025/II/6(b)
	(b) The equation of the line, L, is $2y - x = 12$. It cuts the x-axis and y-axis at Q and P respectively.  (i) Write down the gradient of PQ, [1] (ii) A point K lies on the line L, and is equidistant from x-axis and y-axis. Find the coordinates of K. [2] (iii) Find acute angle QPO. [2]
4	GMSS/2025/I/10
	The diagram in the answer space shows the line $2y = 3x + c$. (a) On the same diagram, sketch and label the line $2y = 3x - 2$.  (b) Find the value c of the line $2y = 3x + c$. [1] (c) Find the equation of another line which is parallel to $2y = 3x + c$ and passes through $(-2, -1)$. [1]

- (b) A straight line, l , with equation $\frac{2x}{a} + \frac{y}{10} = 1$ is parallel to the line $y = 4x$.

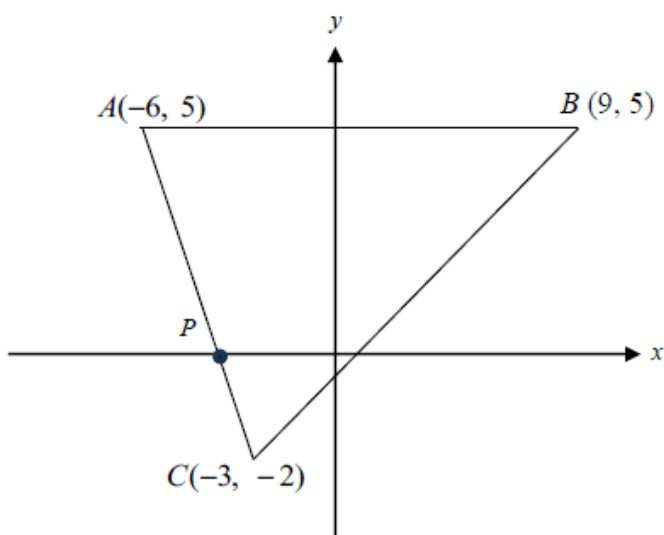


- (i) Show that the value of $a = -5$. [2]
- (ii) State the coordinates of points M and N , where line l intersects the x - and y -axis respectively. [2]
- (c) Line $y = 4x$ intersects the line $y = 10$ at point P .
- (i) Explain why $MNPO$ is a parallelogram. [2]
- (ii) Calculate the area of parallelogram $MNPO$. [2]

6

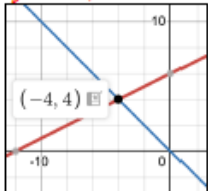
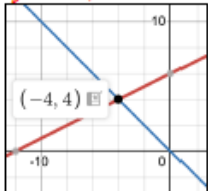
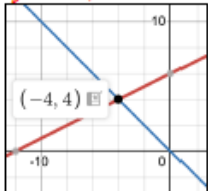
GESS/2025/II/13

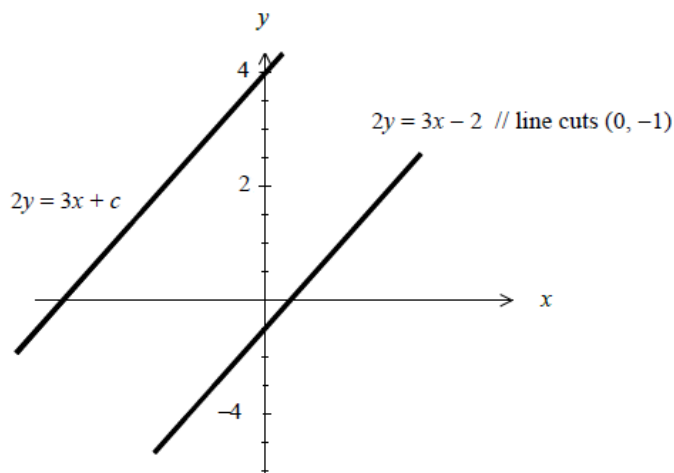
A, B and C are the points $A(-6, 5)$, $B(9, 5)$ and $C(-3, -2)$.



Find

- (a) the equation of line AC , [2]
- (b) the equation of the line that is parallel to AC and passes through B , [2]
- (c) area of triangle ABC , [2]
- (d) the shortest distance from A to BC , [2]
- (e) the coordinate of point P as the line AC intersects the x -axis at P . [2]

1	(a) $\sqrt{(x-5)^2 + (0 - (-2))^2} = 2$ $(x-5)^2 + 4 = 4 \quad \text{--- [M1]}$ $(x-5)^2 = 0$ $x = 5$ S (5, 0) --- [A1] (b) $\text{Gradient of } l_1 = \frac{-10-0}{0-5} = 2 \text{ --- [M1]}$ No, there is a possibility that <u>l_1 and l_2 can possibly be the same line</u> as <u>l_1 and l_2 have the same gradient</u>, <u>they may have the same y-intercept</u>, as y - intercept of l_2 is not given. [A1]		
2	(a) G (3 , 4) (b) DF $= \sqrt{(5-3)^2 + (9-1)^2}$ $= 8.2462$ $= 8.25 \text{ units (3 s.f.)}$		
3	(bi) $2y - x = 12$ $y = \frac{1}{2}x + 6$ Is gradient equivalent to tangent in this context? Gradient of PQ = $\frac{1}{2}$ (bii) 2 possibilities <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 10px; vertical-align: top;"> Since point is equidistant from x-axis and y-axis, therefore $x = -y$. From $2y - x = 12$ $2y + y = 12$ $y = 4, x = -4$  $K = (-4, 4)$ </td><td style="width: 50%; padding: 10px; vertical-align: top;"> Since point is equidistant from x-axis and y-axis, therefore $x = y$. From $2y - x = 12$ $2y - y = 12$ $y = 12, x = 12$  $K = (12, 12)$ </td></tr> </table> (biii) $\tan \angle QPO = \frac{OQ}{OP} = \frac{12}{6} = 2$ acute $\angle QPO = 63.4^\circ$ (to 1 d.p.)	Since point is equidistant from x-axis and y-axis, therefore $x = -y$. From $2y - x = 12$ $2y + y = 12$ $y = 4, x = -4$  $K = (-4, 4)$	Since point is equidistant from x-axis and y-axis, therefore $x = y$. From $2y - x = 12$ $2y - y = 12$ $y = 12, x = 12$  $K = (12, 12)$
Since point is equidistant from x-axis and y-axis, therefore $x = -y$. From $2y - x = 12$ $2y + y = 12$ $y = 4, x = -4$  $K = (-4, 4)$	Since point is equidistant from x-axis and y-axis, therefore $x = y$. From $2y - x = 12$ $2y - y = 12$ $y = 12, x = 12$  $K = (12, 12)$		
4	(a)		



(b)

Sub $x = 0$ and $y = 4$,

$$2(4) = 0 + c$$

$$c = 8$$

(c)

Using $m = 1.5$ and $(-2, -1)$,

$$-1 = 1.5(-2) + c$$

$$c = 2$$

$$y = 1.5x + 2$$

5

(bi) $\frac{6x}{a} + \frac{3y}{10} = 3$

$$60x + 3ay = 30a$$

$$y = -\frac{20}{a}x + 10$$

$$-\frac{20}{a} = 4$$

$$a = -5 \text{ (Shown)}$$

(bii) $y = 4x + 10$

$$M(-2.5, 0)$$

$$N(0, 10)$$

(ci)

MN // OP with gradient of 4.
MO // NP - (M & O share the same y coordinate 0) while (N & P share the same y coordinate 10). They are both parallel horizontal lines.
Since there are two pairs of parallel sides. This is a parallelogram.

(cii) $Area = 10 \times 2.5$
 $= 25$

6

(a)

 $(-6, 5)$, and $(-3, -2)$ *Gradient*

$$\begin{aligned} &= \frac{5 - (-2)}{-6 - (-3)} \\ &= \frac{7}{-3} \end{aligned}$$

$$y = \frac{7}{-3}x + c$$

Sub $(-6, 5)$

$$5 = \frac{7}{-3}(-6) + c$$

$$c = -9$$

$$y = \frac{7}{-3}x - 9$$

(b)

$$y = \frac{7}{-3}x + c$$

Sub $(9, 5)$

$$5 = \frac{7}{-3}(9) + c$$

$$c = 26$$

$$y = \frac{7}{-3}x + 26$$

(c)

$$9 - (-6) = 15$$

$$5 - (-2) = 7$$

$$Area = \frac{1}{2} \times 15 \times 7 = 52.5$$

(d)

 $(9, 5)$ $(-3, -2)$

$$BC = \sqrt{(9 - (-3))^2 + (5 - (-2))^2}$$

$$BC = \sqrt{144 + 49} = \sqrt{193}$$

$$Area = \frac{1}{2} \times d \times \sqrt{193} = 52.5$$

$$d = \frac{52.5}{\frac{1}{2} \times \sqrt{193}}$$

$$= 7.558065$$

$$= 7.56(3sf)$$

(e)

$$y = \frac{7}{-3}x - 9$$

$$\text{Sub } y = 0$$

$$\frac{7}{3}x = -9$$

$$x = \frac{-27}{7}$$

$$P\left(\frac{-27}{7}, 0\right)$$